

Introduction to Distributed System

The background of the slide features a blue gradient that transitions from a deep blue on the left to a lighter, cyan-like blue on the right. Overlaid on this gradient are several smooth, wavy lines in shades of yellow, light blue, and white, creating a sense of movement and depth.

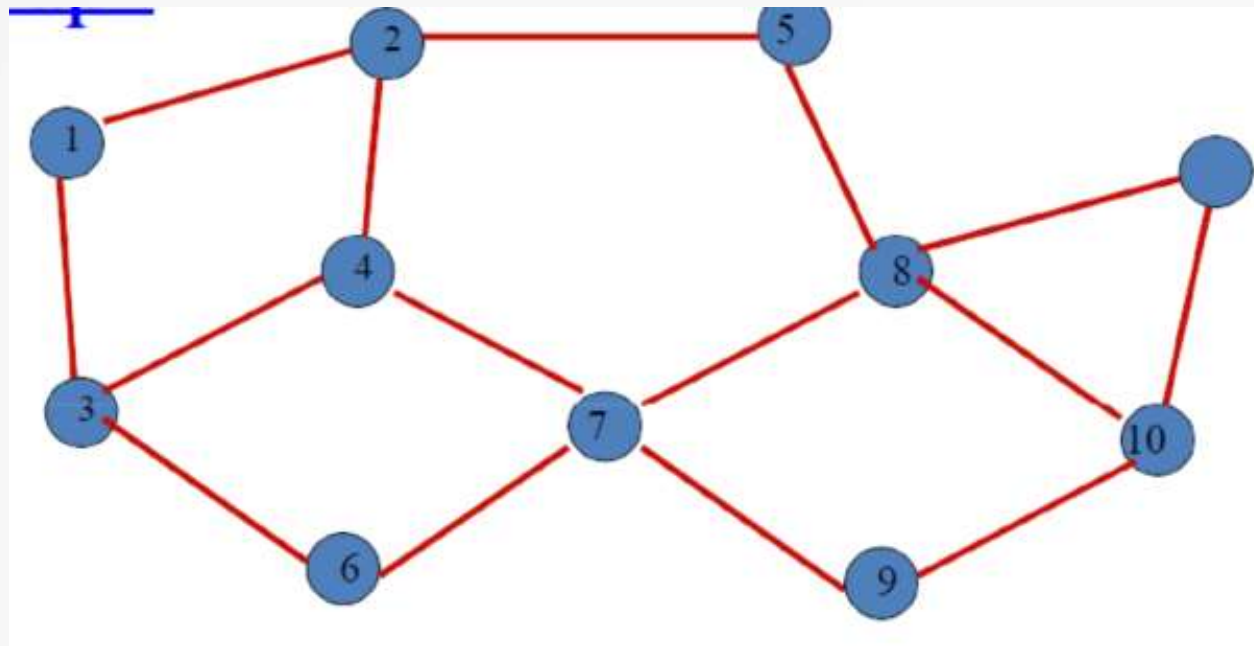
Agenda

- Definition of a Distributed System
 - Typical Distributed System
 - Relation with Shared Memory/Parallel Systems
 - Major Components
 - Interaction with System Components
 - Common Characteristics
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- A decorative graphic on the right side of the slide, consisting of several overlapping, curved, wavy bands in light blue, yellow, and dark blue, creating a modern, abstract design.

Definition of a Distributed System

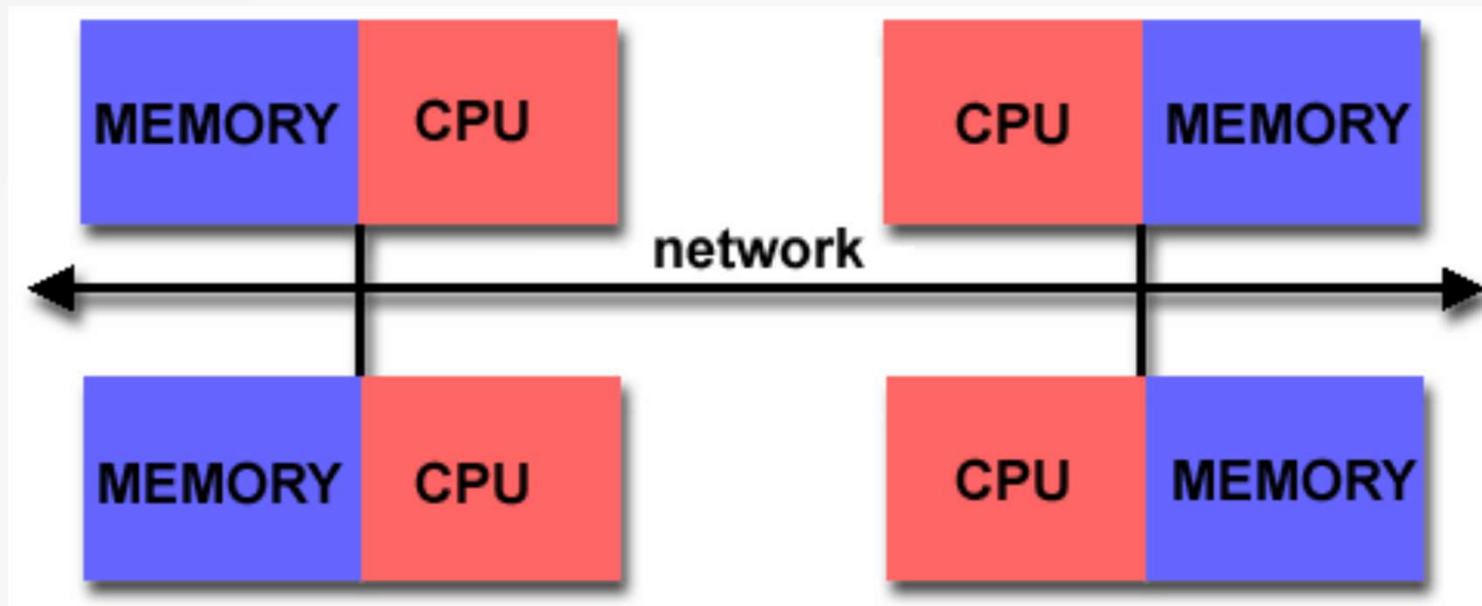
- A collection of independent computers that appears to the users of the system as a **single coherent computer/system image**.
- A collection of computers that do not **share common memory** or a **common physical clock**, that communicate by a messages passing over a communication network, and where each computer has its own memory and runs its own operating system.
- Nodes typically are **loosely coupled**.
- A distributed system is a **collection of independent entities** that cooperate to solve a problem that cannot be individually solved.

Topology of a DS is represented by a graph

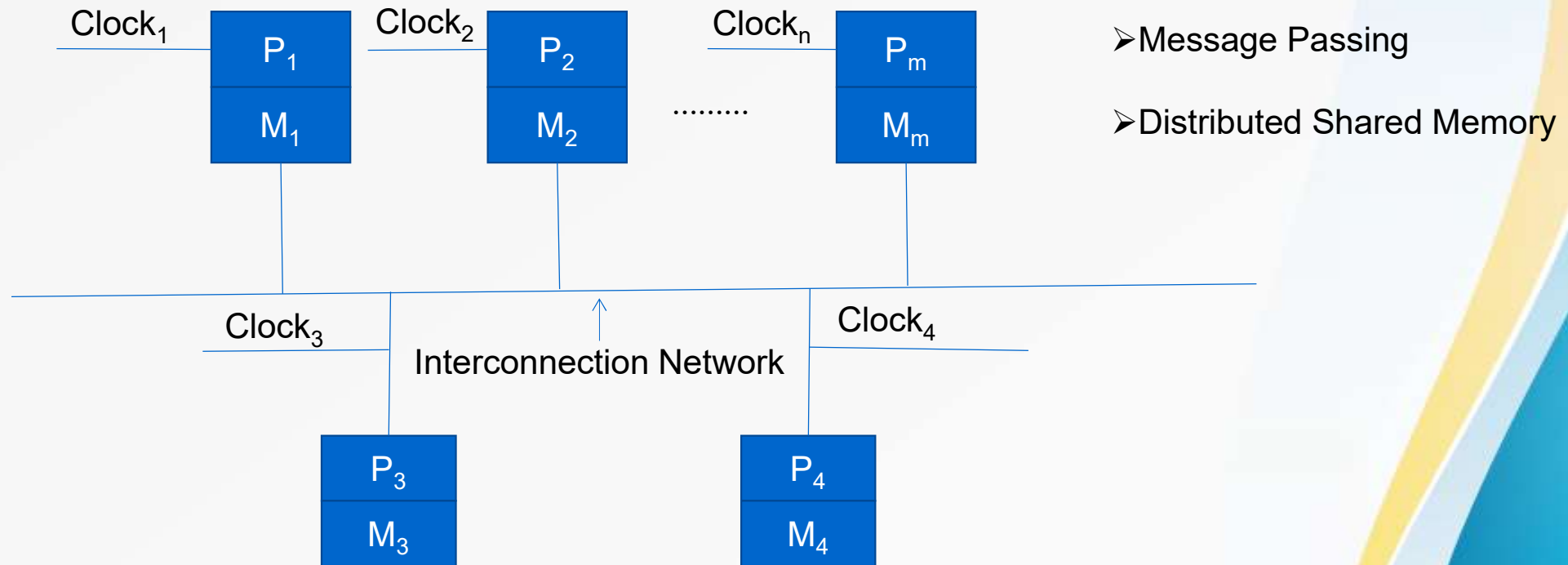


A network of processes. The nodes are processes, and the edges are communication channels.

Typical Distributed System



Another Representation of Distributed Systems



- Nodes may be geographically apart, therefore, there may be longer delays in communication.
- Clock at each of the nodes is different.

Distributed System

➤ Distributed Operating System

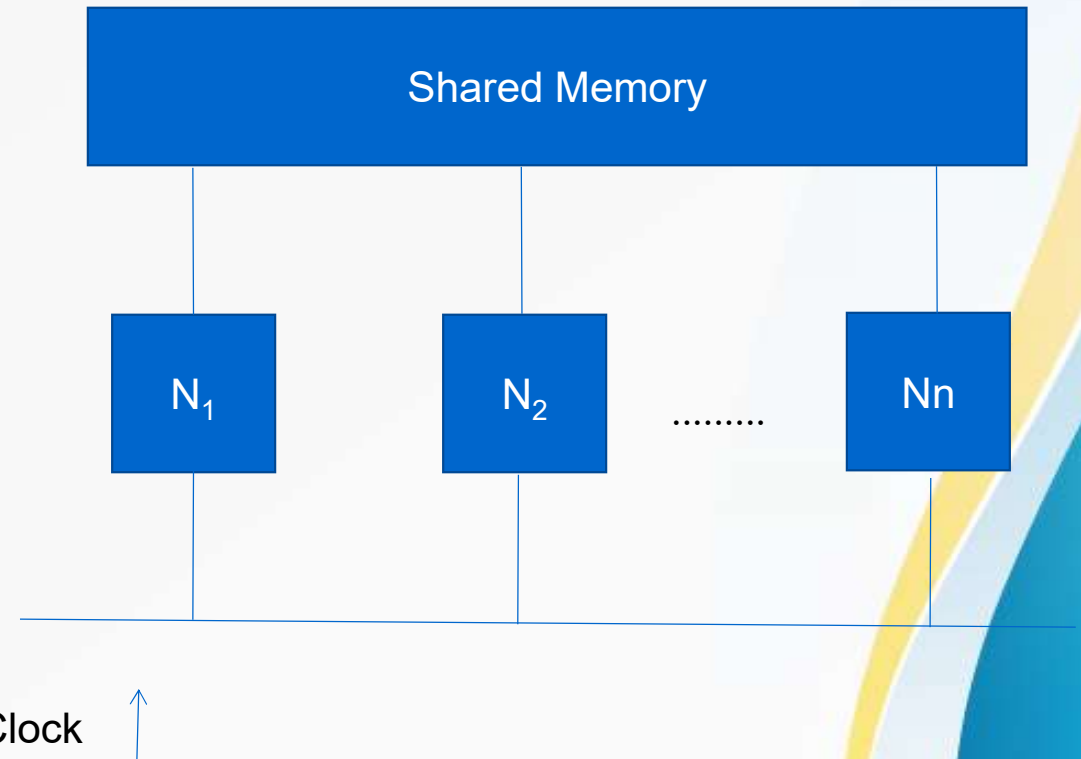
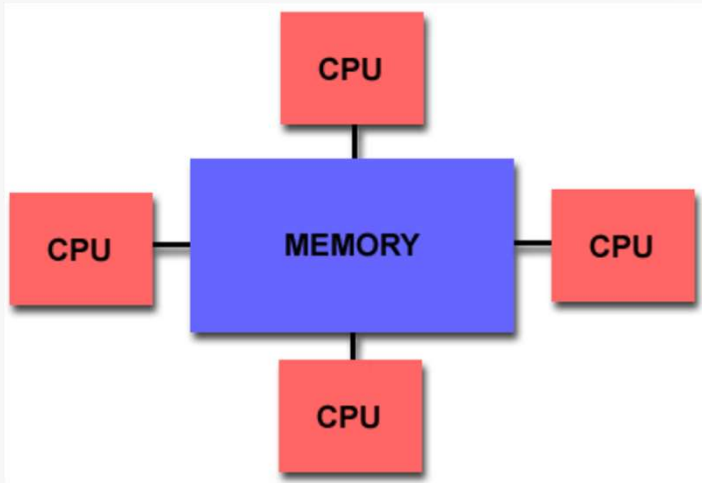
- Identical OS on all the nodes in the system, a set of system calls remain same.

➤ Modes of Communication

- Message Passing- Messages passed between nodes
- Distributed Shared Memory: Physical memories of individual nodes as a single logical space.

Multiprocessor/Parallel Systems

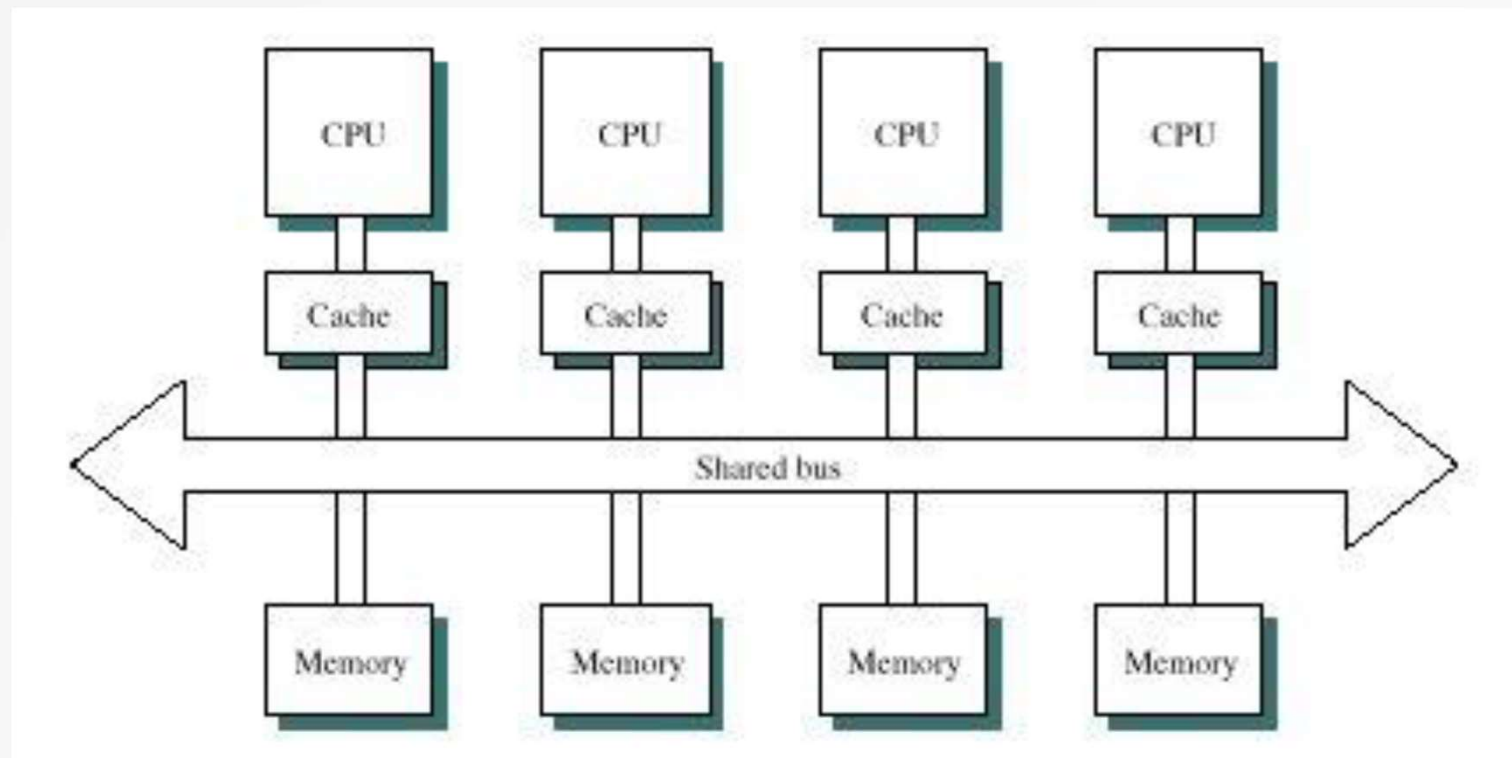
- A multiprocessor system is a parallel system in which multiple processors have direct access to a shared memory which forms a common address space.
- Shared Memory
- Message Passing (distributed)



Shared Memory Computers

- All processors have access to all memory as a global address space
- Multiple processors can operate independently, but share the same memory resources
- Changes in a memory location effected by one processor are visible to all other processors

Symmetric Multiprocessor (SMP)



Parallel Vs Distributed Systems

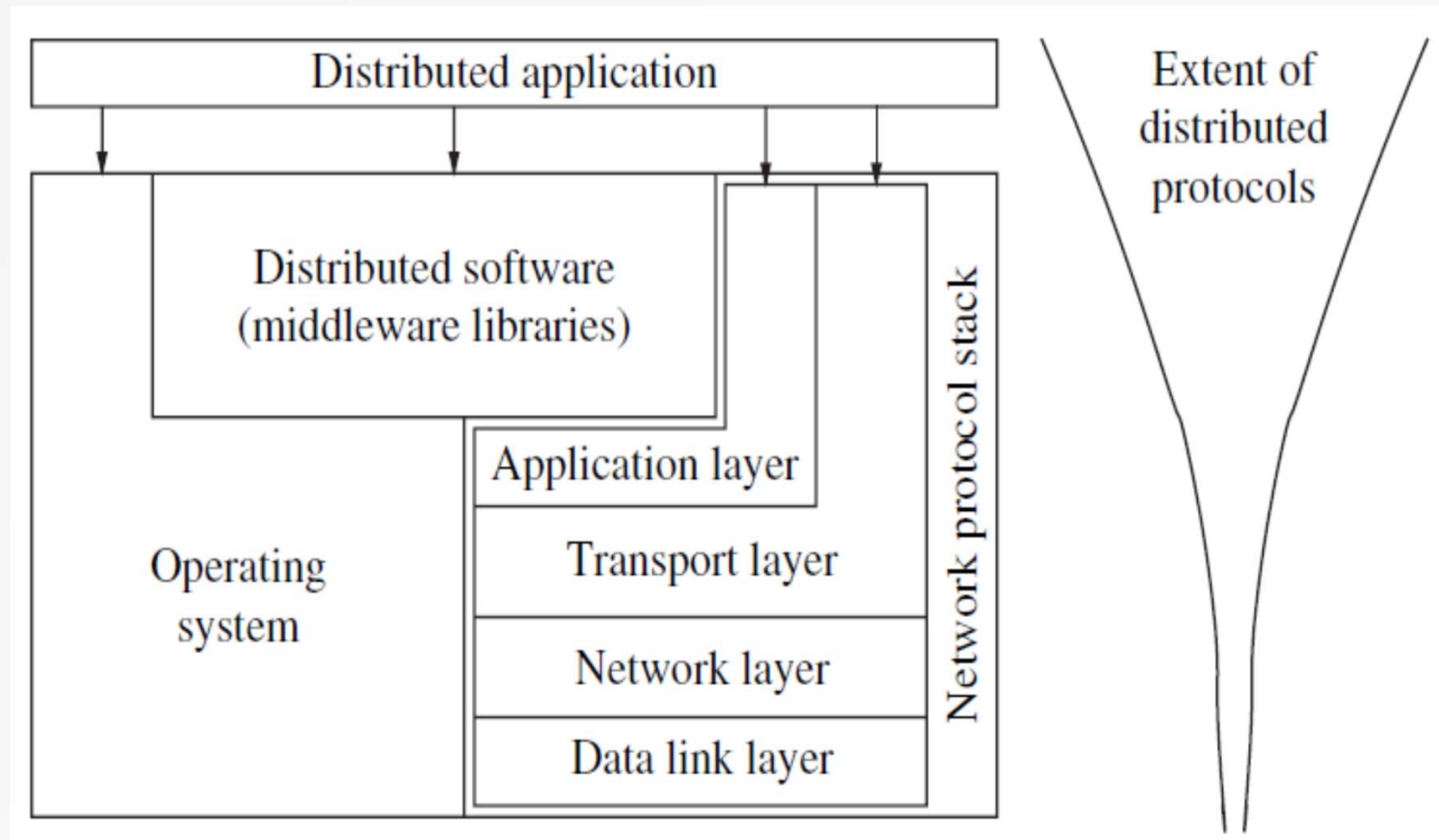
- In parallel systems, the primary issues are
 - speed-up and increased data handling capability.
- In distributed systems the primary issues are
 - fault-tolerance, synchronization, scalability, etc.

	Shared memory	Distributed memory
Programmability	easier	harder
Scalability	harder	easier

Major Components

- Machines/Devices (PCs, Servers, Specialized devices....), commonly called nodes.
- Network (Links, Switches, Routers, etc.)
- Storage (local, distributed)
- Systems software/applications/tools (Distributed OS, databases/filesystems, load balancer, performance monitoring tools, security software...)
- Not all distributed system will need all these.
- Middleware
- User applications

Interaction of the Software Components at each Processor



Ref: Distributed Computing: Principles, Algorithms, and Systems by Ajay D. Kshemkalyani and Mukesh Singhal

Examples

- Large networks are very commonplace these days. Think of the **World Wide Web**.
- Other examples are:
 - Social Networks
 - Sensor networks
 - BitTorrent for downloading video / audio
 - Skype for making free audio and video communication
- Important Services
 - Internet banking
 - Web Search
 - Net meeting
 - Skype

Common Characteristics

- **Heterogeneous in many cases**

- Different architectures, operating systems..

- **Can be geographically distributed**

- Network delays play an important role

- **Faults are common**

- Larger the system is, more chance of something failing at any one time

- **Replication is widely used for better availability and performance** □

- Need for maintaining data consistency
- Trade-off between performance and consistency